Movie & Streaming Content Analytics Dashboard

Business Problem Statement

1. Project Overview

This project analyzes movie and streaming-content data to understand content performance, audience engagement, genre-level patterns, and changes in the catalog over time. The analysis is presented through an interactive Power BI dashboard for quick exploration and comparison.

2. Business Problem

A large content catalog can be difficult to evaluate using raw data alone. This project focuses on turning movie-level information such as ratings, popularity, votes, genres, languages, and release years into clear insights that can support content analysis and decision-making.

3. Project Objectives

- Identify highly rated titles and understand overall rating patterns.
- Compare popularity and average IMDb ratings across genres.
- Examine audience engagement using vote count and vote average.
- Track popularity trends across release years.
- Understand how the amount of content changes over time.
- Provide interactive year and genre filters for detailed exploration.

4. Dataset Overview

Attribute	Details
Total Records	9,837
Main Fields	Release Date, Title, Overview, Popularity, Vote Count, Vote Average, Original Language, Genre, Poster URL

5. Dashboard KPIs

436	10K	6.43	40.32	1.39K
High Rated Titles	Total Content	Average IMDb Rating	Average Popularity	Average Vote Count

6. Key Analysis Areas

Genre Performance: Compares average popularity and average IMDb rating across genres.

Audience Engagement: Uses vote count and vote average to examine audience response.

Content Distribution: Examines content and original-language distribution in the dataset.

Time Trends: Tracks popularity and content growth by release year.

7. Business Questions

- Which genres have the highest average popularity?
- Which genres have the strongest average IMDb ratings?
- How many titles meet the dashboard's high-rating criteria?
- How does content volume change over the years?
- How does average popularity change over the years?
- Is there a visible relationship between vote count and vote average?
- How is content distributed across languages?

8. Tools & Technologies

Python — data preparation and exploratory analysis **SQL** — structured data analysis **Power BI** — interactive dashboard and visualization

9. Expected Outcome

The final dashboard provides a single view of content performance, ratings, popularity, audience engagement, genre patterns, and time-based trends. It allows users to explore the dataset through interactive filters and quickly identify patterns that would be harder to see in raw data.

10. Project Deliverables

Deliverable	Purpose
Business Problem Statement	Defines the project scope and analytical objectives.
Python Notebook	Data preparation and exploratory analysis.
SQL Queries	Structured analysis and business questions.
Power BI Dashboard	Interactive presentation of the analysis.
Project Report	Methodology, findings, and conclusions.
README.md	Project documentation for the GitHub repository.